A One-Stage Domain Adaptation Network With Image Alignment for Unsupervised Nighttime Semantic Segmentation

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Abstract—In this paper, we tackle the problem of semantic segmentation for nighttime images that plays an equally important role as that for daytime images in autonomous driving, but is also much more challenging due to very poor illuminations and scarce annotated datasets. It can be treated as an unsupervised domain adaptation (UDA) problem, i.e., applying other labeled dataset taken in the daytime to guide the network training meanwhile reducing the domain shift, so that the trained model can generalize well to the desired domain of nighttime images. However, current general-purpose UDA approaches are insufficient to address the significant appearance difference between the day and night domains. To overcome such a large domain gap, we propose a novel domain adaptation network "DANIA" for nighttime semantic image segmentation by leveraging a labeled daytime dataset (the source domain) and an unlabeled dataset that contains coarsely aligned day-night image pairs (the target daytime and nighttime domains). These three domains are used to perform a multi-target adaptation via adversarial training in the network. Specifically, for the unlabeled day-night image pairs, we use the pixel-level predictions of static object categories on a daytime image as a pseudo supervision to segment its counterpart nighttime image. We also include a step of image alignment to relieve the inaccuracy caused by the misalignment between day-night image pairs by estimating a flow to refine the pseudo supervision produced by daytime images. Finally, a re-weighting strategy is applied to further improve the predictions, especially boosting the prediction accuracy of small objects. The proposed DANIA is a one-stage adaptation framework for nighttime semantic segmentation, which does not train additional day-night image transfer models as a separate pre-processing stage. Extensive experiments on Dark Zurich and Nighttime Driving datasets show that our DANIA achieves state-of-the-art performance for nighttime semantic segmentation.

Index Terms—Domain adaptation, semantic segmentation, nighttime, image alignment

1 Introduction

Aiming to label each pixel of a given image to an object category, semantic segmentation is a fundamental computer vision task and benefits many applications such as autonomous driving [1], [2], medical imaging [3] and human parsing [4]. With the advancement of deep learning techniques and computing power, the state-of-the-art performance of semantic segmentation for natural scene images taken at the daytime with favorable illumination conditions has been significantly improved in recent years [5], [6], [7]. It is noteworthy that this performance improvement is credited to the large amount of pixel-level annotated images. However, the performance of segmenting images that are taken in adverse weather/lighting conditions such as foggy weather [8] and nighttime [9], [10], [11] is still not satisfactory due to the lack of well-annotated datasets for training. With many indiscernible regions and visual hazards [12], e.g., under/over exposure and motion blur, it is much harder to manually build the pixel-level annotations for those realistic adverse weather or nighttime scene images. In this paper, we focus on semantic image segmentation in the nighttime, which is as frequent and important as that in the daytime, especially to autonomous driving.

Directly applying the networks trained on daytime datasets to segment nighttime images suffers from serious data bias [11] and may result in 50% accuracy drop, as shown in our later experiments (e.g., Table 1). Many unsupervised domain adaptation (UDA) approaches have been proposed to address such data bias by bridging the gap between a labeled domain and an unlabeled domain [13], [14], [15]. Taking daytime and nighttime as the labeled and unlabeled domains respectively, UDA can provide a general solution to the nighttime semantic segmentation through feature-distribution alignment or adversarial training. Many general-purpose UDA approaches are designed under the setting of synthetic-to-real [16], [17] – using a labeled synthetic dataset and an unlabeled real dataset to alleviate the burden of large-scale real-data manual annotation. However, this setting is not applicable to the nighttime semantic segmentation due to the lack of large-scale labeled synthetic nighttime image dataset.

Given the large appearance (e.g., illumination) difference between daytime and nighttime, it is also difficult to use

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TABLE 1
The mIoU (%) Performance of the Pre-Trained Semantic Segmentation Models on the Validation Set of Cityscapes and Dark Zurich

Method	Cityscapes-val	Dark Zurich-val
RefineNet [59]	65.20	15.16
DeepLab-v2 [58]	65.67	12.14
PSPNet [60]	63.37	12.28

general-purpose UDA approaches to adapt the network training directly from the (labeled) daytime domain to the (unlabeled) nighttime domain, as shown in our later experiments (e.g., Table 2). To address this problem, several variants of UDA approaches have been proposed to transfer the semantic segmentation models from daytime to nighttime without using labels in the nighttime domain. In [9], [18], [19], an intermediate twilight domain is introduced as a bridge to build a smooth adaptation from daytime to nighttime. In [9], [11], [19], [20], [21], [22], an image transferring network is trained to stylize nighttime or daytime images and construct synthetic datasets for domain adaptation. However, these two lines of works suffer from two major limitations:

1) Negative transfer. All these methods require an additional pre-processing stage of training an image-transfer model between daytime and nighttime. This is not only time-consuming but also making the performance of later stages to be highly dependent on the performance of the pre-processing stage. Especially, there is no guarantee that the transferred pixels can keep exactly the same semantic information as in the original images when the domain gap is large [23].
2) Usage of intermediate domain. Some of the above approaches take twilight as an intermediate domain between day and night to perform a multi-stage adaptation which introduces more computational burden. What's more, compared to day and night, twilight scene images are more difficult to capture due to its strict definition according to the solar elevation angle [18].

To address the above limitations, we propose in this paper a novel one-stage *D*omain *A*daptation *N*etwork with *I*mage *A*lignment "DANIA" for nighttime semantic segmentation by taking advantage of the newly released Dark Zurich dataset [9], [19], which contains unlabeled day-night scene image pairs that are coarsely aligned using GPS recordings. The proposed DANIA performs a multi-target adaptation from Cityscapes data (the source domain) to Dark Zurich daytime (Dark Zurich-D, the target daytime domain) and nighttime data (Dark Zurich-N, the target nighttime domain). Specifically, we first adapt the model from Cityscapes, which contains large-scale training images with labels, to Dark Zurich-D since they are all taken at daytime. Then, the prediction of Dark Zurich-D is used as a pseudo supervision for Dark Zurich-N for network training. We apply an image relighting subnetwork to make the intensity distribution of the images from different domains to be close. Following [13], we apply a weight-sharing semantic segmentation network to make predictions for the relighted images and perform an adversarial learning in the output space to ensure very close layout across different

TABLE 2
The Per-Category mIoU (%) on Dark Zurich-Test by Current State-of-the-Art Methods and Our DANIA

Method	road	sidewalk	building	wall	fence	pole	traffic light	traffic sign	vegetation	terrain	sky	person	rider	car	truck	bus	train	motorcycle	bicycle	**mIoU**
RefineNet [59]-Cityscapes	68.8	23.2	46.8	20.8	12.6	29.8	30.4	26.9	43.1	14.3	0.3	36.9	49.7	63.6	6.8	0.2	24.0	33.6	9.3	28.5
DeepLab-v2 [58]-Cityscapes	79.0	21.8	53.0	13.3	11.2	22.5	20.2	22.1	43.5	10.4	18.0	37.4	33.8	64.1	6.4	0.0	52.3	30.4	7.4	28.8
PSPNet [60]-Cityscapes	78.2	19.0	51.2	15.5	10.6	30.3	28.9	22.0	56.7	13.3	20.8	38.2	21.8	52.1	1.6	0.0	53.2	23.2	10.7	28.8
AdaptSegNet-Cityscapes→DZ-night [13]	86.1	44.2	55.1	22.2	4.8	21.1	5.6	16.7	37.2	8.4	1.2	35.9	26.7	68.2	45.1	0.0	50.1	33.9	15.6	30.4
ADVENT-Cityscapes→DZ-night [78]	85.8	37.9	55.5	27.7	14.5	23.1	14.0	21.1	32.1	8.7	2.0	39.9	16.6	64.0	13.8	0.0	58.8	28.5	20.7	29.7
BDL-Cityscapes→DZ-night [14]	85.3	41.1	61.9	32.7	17.4	20.6	11.4	21.3	29.4	8.9	1.1	37.4	22.1	63.2	28.2	0.0	47.7	39.4	15.7	30.8
IntraDA-Cityscapes→DZ-night [79]	83.3	28.6	55.0	17.4	14.0	29.3	21.2	23.0	36.5	13.1	0.3	35.1	21.3	63.6	39.8	0.0	26.5	23.5	13.9	28.7
FDA-Cityscapes→DZ-night [38]	84.3	41.5	51.0	12.4	10.1	36.3	32.0	35.3	49.9	10.3	15.7	37.6	43.1	60.9	5.9	0.0	7.5	29.9	13.0	30.3
DMAda [18]	75.5	29.1	48.6	21.3	14.3	34.3	36.8	29.9	49.4	13.8	0.4	43.3	50.2	69.4	18.4	0.0	27.6	34.9	11.9	32.1
GCMA [9]	81.7	46.9	58.8	22.0	20.0	41.2	**40.5**	**41.6**	64.8	31.0	32.1	**53.5**	47.5	**75.5**	39.2	0.0	49.6	30.7	21.0	42.0
MGCDA [19]	80.3	49.3	66.2	7.8	11.0	**41.4**	38.9	39.0	64.1	18.0	55.8	52.1	**53.5**	74.7	**66.0**	0.0	37.5	29.1	22.7	42.5
DANNet (DeepLab-v2) [24]	88.6	53.4	69.8	34.0	20.0	25.0	31.5	35.9	69.5	32.2	82.3	44.2	43.7	54.1	22.0	0.1	40.9	36.0	24.1	42.5
DANNet (RefineNet) [24]	90.0	54.0	**74.8**	**41.0**	21.1	25.0	26.8	30.2	72.0	26.2	84.0	47.0	33.9	68.2	19.0	**0.3**	66.4	38.3	23.6	44.3
DANNet (PSPNet) [24]	90.4	60.1	71.0	33.6	22.9	30.6	34.3	33.7	70.5	31.8	80.2	45.7	41.6	67.4	16.8	0.0	**73.0**	31.6	22.9	45.2
DANIA (DeepLab-v2)	89.4	60.6	72.3	34.5	23.7	37.3	32.8	40.0	**72.1**	33.0	84.1	44.7	48.9	59.0	9.8	0.1	40.1	38.4	**30.5**	44.8
DANIA (RefineNet)	90.8	59.7	73.7	39.9	**26.3**	36.7	33.8	32.4	70.5	32.1	**85.1**	43.0	42.2	72.8	13.4	0.0	71.6	**48.9**	23.9	**47.2**
DANIA (PSPNet)	**91.5**	**62.7**	73.9	39.9	25.7	36.5	35.7	36.2	71.4	**35.3**	82.2	48.0	44.9	73.7	11.3	0.1	64.3	36.7	22.7	47.0

Cityscapes→DZ-night denotes the adaptation from Cityscapes to Dark Zurich-night. The best results are presented in ***bold****, with the second best results underlined.*

domains. To relieve the misalignment caused by the rough GPS recording, we incorporate an image alignment strategy to estimate the flow underlying the day-night image pairs and then use it to rectify the pseudo labels provided by daytime images. In addition, a re-weighting strategy is proposed to boost the prediction accuracy of small objects. We conduct extensive experiments on Dark Zurich and Nighttime Driving datasets to justify the effectiveness of the proposed DANIA for nighttime semantic segmentation.

Preliminary results of this work have been published in a conference proceedings [24]. Compared with the published conference paper, this article is rewritten with many new contents and the following extensions:

- The proposed method is further enhanced by incorporating image alignment to relieve the misalignment of the day-night image pairs and all the experiments are also updated to reflect the performance gains over the preliminary ones in the conference version.
- More extensive comparison results are provided, including comparison with more recent state-of-the-art UDA approaches for the application of semantic segmentation, ablation studies with the new features introduced in this work, and the scenario of using day-night pairs in the testing stage.
- More analytical results and discussions are added, e.g., a histogram visualization of the intensity distributions and the usage of various image pre-processing methods in the proposed method.

2 Related Work

2.1 Domain Adaptation for Semantic Segmentation

Domain adaptation methods are developed to transfer knowledge learned from a source domain to a target domain, which shares similar objects yet different data distributions, so that a comparable prediction of the target domain can be obtained without its own supervision. Recently, many UDA approaches are proposed for the application of semantic segmentation to avoid the high costs of manual dense annotation. In [25], Hoffman *et al.* proposed a novel fully convolutional domain adversarial learning approach with category constraints [26] for semantic segmentation. Tsai *et al.* [13] later developed a multi-level adversarial network to perform domain adaptation in the output space. Luo *et al.* [27] further improved the global adversarial alignment by considering the category-level joint distribution alignment.

Other than adversarial learning techniques, image translation and style transfer [28] from source images to target ones, or vice versa, have been widely used for domain adaptation [29], [30]. In [31], [32], [33], domain-invariant representations are obtained in the process of image translation between the source and target domains. Another line of researches [15], [34] adopted the curriculum-style learning by first learning easy properties in the target domain and then using it to regularize the semantic segmentation model. Recently, self-training also has become a common strategy to boost the performance by iteratively predicting and fine-tuning a set of pseudo labels in multiple rounds of network trainings, as shown in [14], [35], [36], [37], [38].

All the above methods focus on the domain adaptation for synthetic-to-real (i.e., GTA5 [16] or SYNTHIA [17] to Cityscapes) or cross-city images (i.e., Cityscapes to Cross-City [39]), which are all daytime to daytime adaptations. In this paper, we instead focus on the adaptation between the daytime and the nighttime domains with significantly different illumination patterns [9]. These general-purpose domain adaptation approaches could not handle well the significant domain gap between the daytime and nighttime images for satisfactory performance in nighttime semantic segmentation [9].

2.2 Nighttime Semantic Segmentation

As a subproblem of UDA, existing nighttime semantic segmentation methods stem from the above UDA approaches. Dai *et al.* [18] first leveraged an intermediate twilight domain to progressively adapt semantic models trained in daytime scenes to nighttime ones. Sakaridis *et al.* [9], [19] further extended it to a guided curriculum adaptation framework, which uses both the stylized synthetic images and the unlabeled real images to exploit the cross-time-of-day correspondence of the scene images. However, such gradual adaptation approaches usually need to train multiple semantic segmentation models, e.g., three models in [9] for three different domains respectively, which is highly inefficient. Following works along this line [20], [21], [22] also train some additional image transfer models, e.g., CycleGAN [28], to perform the day-to-night or night-to-day image transfer before training the semantic segmentation models. For these methods, the performance of adaptation and semantic segmentation is highly dependent on the image transfer model pre-trained in the pre-processing stage.

Vertens *et al.* [40] proposed to leverage the thermal infrared images as a complementary input to the RGB images for nighttime semantic segmentation since thermal radiation is not very sensitive to the illumination changes. In [10], a two-stage adversarial training method was proposed for semantic segmentation of rainy night scenes by performing domain adaptation between day-night near scene pairs. More recently, Zheng *et al.*[11] proposed a task-agnostic image translation approach which decouples the domain-specific and domain-invariant information to boost performance of semantic segmentation and object detection in adverse weathers.

Different from all the above methods, the proposed DANIA in this paper performs a one-stage end-to-end adversarial learning for training the nighttime semantic segmentation network without using any other image modality.

2.3 Image Alignment

Image alignment aims to establish the spatial correspondence between a pair of images. It is a fundamental problem in computer vision and benefits several important applications including visual localization [41], [42], [43], two-view geometry estimation [44], 3D reconstruction [45] and optical flow estimation [46], [47]. Existing image alignment techniques basically can be divided into two categories, parametric and non-parametric ones [46], based on the representation of the correspondence between the image pairs. Typical parametric approaches use RANSAC

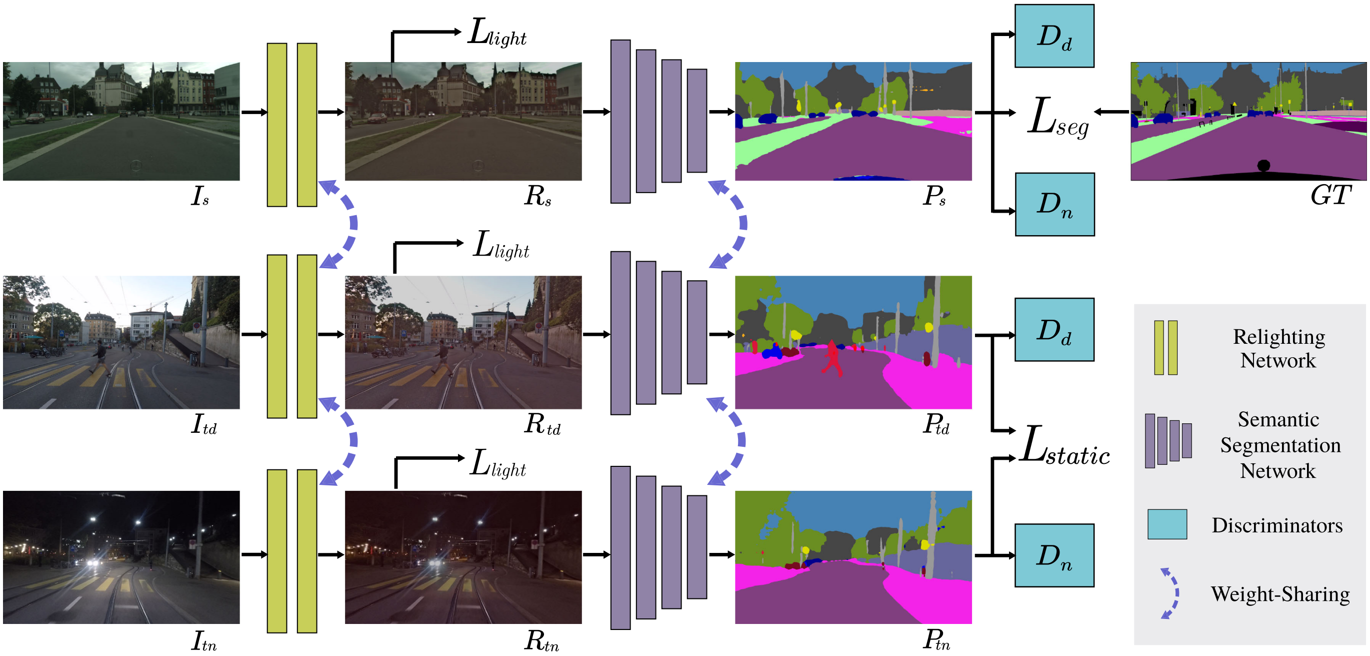


Fig. 1. The architecture of the proposed DANIA. Three input images I_s, I_{td}, and I_{tn} are drawn from the source domain S (Cityscapes) and two target domains T_d and T_n (Dark Zurich-D and Dark Zurich-N), respectively. They go through a weight-sharing image relighting network which can make their distributions be close to each other via the light loss L_{light}. All the outputs are fed into a weight-sharing segmentation network to obtain the predictions of the three domains, P_s, P_{td} and P_{tn}, respectively. For the prediction from I_s, a semantic segmentation loss L_{seg} is computed using the ground-truth segmentation from the source dataset. Besides, the prediction from I_{td} for the categories of static objects provide weak supervision for the corresponding categories from I_{tn}, reflected by a static loss L_{static}. Note that the composition of the relighting network and the semantic segmentation network forms the generator G. Two discriminators D_d and D_n are designed to distinguish outputs from the source domain S or the target domain T_d and from the source domain S or the target domain T_n, respectively. Note that the gradients from the static loss flow via only P_{tn}^S, the predictions of static object categories, and the daytime prediction is detached as the label when calculating the static loss.

[48] to estimate a transformation matrix from sparse image features such as SIFT [49] and LIFT [50]. Non-parametric approaches instead aim to achieve dense correspondence (e.g., optical flow and disparity map) between two images optimized by pixel-level consistency constraints [51], [52]. There are also few hybrid methods that perform both parametric and non-parametric approaches [46], [53], [54], [55], [56] for image alignment.

In this paper, the day-night image pairs are taken with roughly aligned GPS parameters. We found that the offset of the corresponding points across such a day-night image pair can be in the order of hundreds of pixels, which may affect the effectiveness of the proposed UDA algorithm. To solve this problem, we integrate a most recent image-alignment technique [46] to our method to better align the day-night pairs and further improve the performance of the nighttime semantic segmentation.

3 Proposed Method

3.1 Framework Overview

Our method involves a source domain S and two target domains T_d and T_n, where S, T_d, and T_n represent Cityscapes (daytime), Dark Zurich-D (daytime), and Dark Zurich-N (nighttime), respectively. Note that only the source domain S of Cityscapes has ground-truth semantic segmentation in training. The proposed DANIA proceeds the domain adaptation from S to T_d and S to T_n simultaneously and it consists of three different modules: an image relighting network, a semantic segmentation network, and two discriminators, as illustrated in Fig. 1.

Note that $\mathcal{T}_d$ in our method is treated as an intermediate domain between $\mathcal{S}$ and $\mathcal{T}_n$ — $\mathcal{T}_d$ shares the illumination similarity with $\mathcal{S}$ and content similarity with $\mathcal{T}_n$. Our method works with the assumption that the domain gap between $\mathcal{S}$ and $\mathcal{T}_d$ is less than the gap between $\mathcal{S}$ and $\mathcal{T}_n$, and therefore, the daytime branch can help the nighttime branch.

3.2 Network Architecture

All modules of the proposed DANIA are elaborated in detail below.

Image Relighting Network. Inspired by [57], we design an image relighting network to make the intensity distributions of the images from different domains to be close to each other so that the later semantic segmentation network is less sensitive to illumination changes. The relighting network takes the scene images I_s, I_{td} and I_{tn} from the three domains as inputs, and generates the relighted images R_s, R_{td} and R_{tn}, respectively. The relighting network shares weights for all input images from the three domains, see Fig. 2 for the detailed structure of this network.

Semantic Segmentation Network. We select three widely-used semantic segmentation networks to be incorporated into our method: Deeplab-v2 [58], RefineNet [59] and PSPNet [60]. Note that the common backbone is ResNet-101 [61] in all of them. For this module, we share weights for all the input images from the three domains. The semantic segmentation network takes R_s, R_{td} and R_{tn} as the inputs and produces segmentation predictions (category-likelihood map) P_s, P_{td} and P_{tn} for the three domains, respectively. The composition of the image relighting network

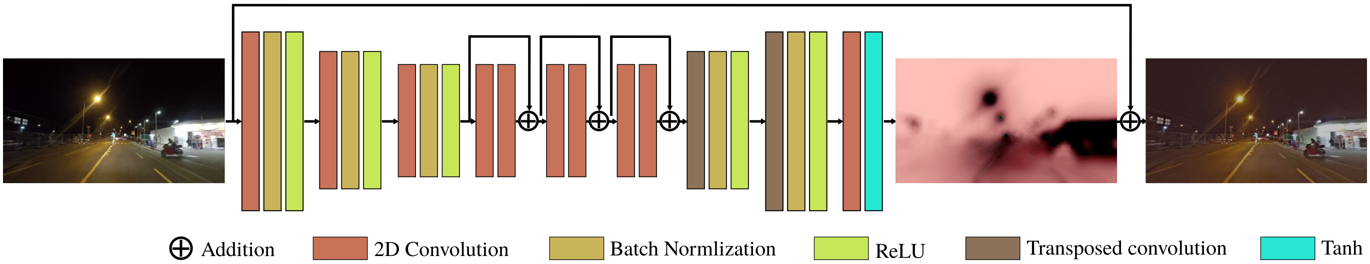


Fig. 2. The structure of the image relighting network. It consists of four convolutional layers, three residual blocks and two transposed convolutional layers, and each convolutional layer is followed by a batch normalization layer except for the last convolutional layer. The output from the last layer is then added to the input image to obtain the relighted image.

and the semantic segmentation network forms the generator G of the proposed DANIA.

Discriminators. As in [13], the discriminators are designed to distinguish whether the segmentation prediction comes from the source domain or either of the target domains by performing adversarial learning in the output space. Following [13], we modified the architecture in [62] by utilizing all fully convolutional layers. Particularly, it includes 5 convolutional layers with the channel numbers of $\{64, 128, 256, 256, 1\}$, and a kernel size of 4×4. The stride is 2 for the first two convolutional layers and 1 for the rest. Since we have two target domains T_d and T_n, we design two discriminators D_d and D_n to distinguish whether the output is from S or T_d and from S or T_n, respectively. The two discriminators share the same structures and the weights are jointly trained.

3.3 Probability Re-Weighting

Due to the fact that the numbers of pixels for different object categories are imbalanced in the source domain, network training can usually converge more easily by predicting a pixel to be a category of large-size object, such as road, building, and tree, in training discriminators. In this case, it is quite difficult to correctly predict the pixels of small objects which have relatively fewer annotations in the dataset, such as pole, sign, and light via a global alignment in the output space. To address this problem, we propose a reweighting strategy to adjust the predicted category-likelihood maps. Specifically, for each category $k \in \mathbb{C}$, we first define a weight

$$w'_k = -\log{(a_k)}, \tag{1}$$

where a_k is the proportion of all the valid pixels that are labeled as category k in the source domain. Clearly the smaller the value of a_k, the larger the value of w'_k and the use of such a weight can help segment the categories of smaller-size objects. Here, the logarithm is computed to prevent from overweighting small-size object categories. In our experiment, we further normalize this weight by

$$w_k = \frac{w'_k - \overline{w}}{\sigma(w)} \cdot std + avg, \tag{2}$$

where $\overline{w}$ and $\sigma(w)$ are the mean and standard deviation of $w'_k, k \in \mathbb{C}$, respectively. The parameters std and avg are two positive constants we pre-select to shift the value range of w_k to be mainly positive. During training, we set $std = 0.1$ and $avg = 1.0$ empirically. We then multiply each normalized weight w_k with the corresponding category channel of the predicted likelihood map P, where $P \in \{P_{td}, P_{tn}\}$. Thus, the final semantic segmentation result F is obtained by employing an argmax operation on the multiplication result.

3.4 Day-Night Correspondence

As shown in Fig. 1, there exists a large number of pixel shift between P_{td} and P_{tn} due to view changes and camera motions between the daytime and nighttime images. This non-negligible misalignment between the day-night pairs limits the reliable prediction reference that the daytime image can provide to its nighttime counterpart. Therefore, we integrate an image alignment approach [46] to establish dense correspondence between the daytime image I_{td} and the nighttime image I_{tn} to avoid the incredible pseudo labels.

Following [46], a ResNet-50 pretrained on ImageNet is applied to extract the multi-level deep feature from I_{td} and I_{tn} separately. Then the RANSAC algorithm is employed to estimate a homography which can be seen as a coarse flow between the two images. After that, a fine-flow network [46] pretrained on MegaDepth [63] dataset is trained to refine the coarse flow $\mathcal{F}_{coarse}$, resulting in a fine-flow map $\mathcal{F}_{fine}$. Finally, the estimated flow $\mathcal{F}_{fine}$ is used to warp the semantic segmentation prediction P_{td} to obtain P'_{td}, as illustrated in Fig. 3.

3.5 Objective Functions

In this subsection, we introduce all the objective functions involved in the proposed end-to-end DANIA training, including the light loss, the semantic segmentation loss, the static loss, and the adversarial loss.

3.5.1 Light Loss

The light loss is proposed to ensure that the intensity distributions of the outputs R_s, R_{td} and R_{tn} after the image relighting network are close to each other. The light loss is a combination of four loss functions: the total variation loss L_{tv}, the exposure control loss L_{exp}, the structural similarity loss L_{ssim}, and style transfer loss L_{style}.

The total variation loss L_{tv} [64] is widely used in image denoising [65] and image synthesis [66] to make images smoother. In this paper, we apply such a loss function to remove rough textures, such as noises, to facilitate the semantic segmentation. The loss L_{tv} is defined by

$$L_{tv} = \frac{1}{N}\|(\nabla_x(I-R))^2 + (\nabla_y(I-R))^2\|_1, \tag{3}$$

where $I \in \{I_s, I_{td}, I_{tn}\}$ represents the input images, $R \in \{R_s, R_{td}, R_{tn}\}$ is the output of the relighting network, N is

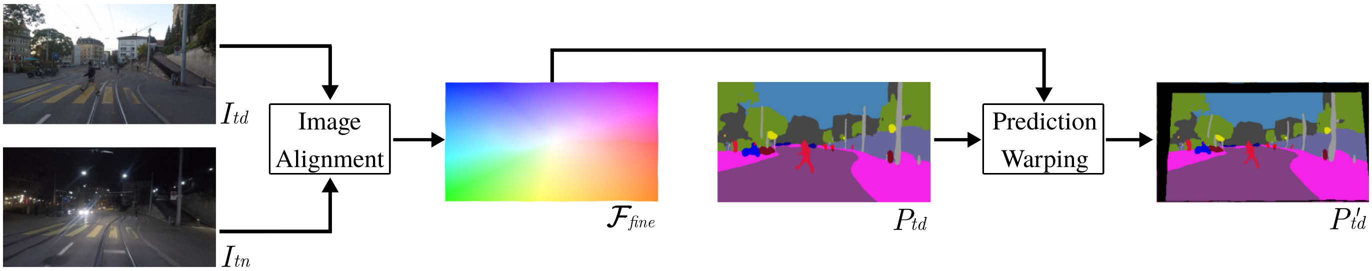


Fig. 3. The process of building the correspondence between the day-night image pairs and prediction warping. The estimated flow in this sample indicates a zoom in and out effect between the image pairs.

the number of pixels in I, ∇_x and ∇_y represent intensity gradients between neighboring pixels along the x and y directions, respectively, and $\|\cdot\|_1$ is the L_1 norm that sums up over all the pixels.

To obtain the similar lighting effects in the day and night scenarios, we apply the following exposure loss L_{exp} proposed in [67] to control the exposure level

$$L_{exp} = \frac{1}{M}\|\varphi(R) - E\|_1, \tag{4}$$

where φ is a 32×32 average pooling function and M represents the number of pixels in $\varphi(R)$.

According to [67], the exposure loss minimizes the distance between the average intensity of the local regions from the low-light images and a pre-defined well-exposed level E to adjust the exposure level of the low-light images. Instead, we leverage this loss function here to adjust the exposure level of the daytime images based on the intensity of the nighttime images. Therefore, different from [67], all pixels on E here have the same value which are dynamically set to be the average intensity value of the nighttime images in this training batch and this value changes in each iteration. For the same purpose, we also compute the Gram matrices [68] which has remarkable capability to capture visual styles and following [69] calculate the style loss L_{style}

$$L_{style} = \frac{1}{N}\|G_R - G_{I_{tn}}\|_2, \tag{5}$$

where G_R and $G_{I_{tn}}$ are the Gram matrices of the relighted image R and the input nighttime image I_{tn}.

The structural similarity loss L_{ssim} [70] is widely used for image reconstruction [71], [72]. Here we apply this loss function to ensure that the generated relighted images R could maintain the structure of original images I. The loss L_{ssim} is defined by

$$L_{ssim} = \frac{1}{2N}\|1 - SSIM(I, R)\|_1. \tag{6}$$

As in [71], we use a simplified SSIM (structural similarity index measure) with a 3×3 block filter in this loss function.

Finally, by combining all the four loss terms, our light loss L_{light} is defined by

$$L_{light} = \alpha_{tv}L_{tv} + \alpha_{exp}L_{exp} + \alpha_{style}L_{style} + \alpha_{ssim}L_{ssim}, \tag{7}$$

where $\alpha_{tv}, \alpha_{exp}$, α_{style}, and α_{ssim} are set to 10, 1, 10 and 1 respectively in all experiments.

3.5.2 Semantic Segmentation Loss

We adopt the widely used weighted cross-entropy loss for training the semantic image segmentation in the source domain

$$L_{seg} = -\frac{1}{N|\mathbb{C}|}\sum_{k\in\mathbb{C}}\|w_k GT^{(k)} \cdot \log(P_s^{(k)})\|_1, \tag{8}$$

where $P_s^{(k)}$ is the kth channel of the prediction P_s from the source images, w_k is the weight defined in Eq. (2), and $GT^{(k)}$ is the one-hot encoding of the ground truth for the kth category.

3.5.3 Static Loss

Based on the fact that a daytime image shares semantic similarities with its corresponding nighttime counterpart when considering only the static object categories, we here introduce a static loss to provide pixel-level pseudo supervision for the static object categories, e.g., road, sidewalk, wall, fence, pole, light, sign, vegetation, terrain and sky, in the nighttime images.

More formally, given the segmentation predictions $P_{tn} \in \mathbb{R}^{H\times W\times C}$, we only consider the channels corresponding to the static categories for calculating this loss. Let's denote $C^{\mathcal{S}}$ as the total number of the categories of static objects, then it holds that $P_{tn}^{\mathcal{S}} \in \mathbb{R}^{H\times W\times C^{\mathcal{S}}}$.

To reduce the wrong supervision provided by the P_{td} because of the misalignment, we use the fine flow $\mathcal{F}_{fine}$ estimated from the day-night image pair to warp the P_{td}. Finally, the refined daytime prediction can be obtained by

$$P'_{td} = \mathcal{F}_{fine}(P_{td}). \tag{9}$$

The global image alignment can help to rectify the prediction of the daytime image, especially for the large-size objects. However, it might hurt the prediction of the small objects and pixels around the boundaries of the large objects, i.e., a pole may be warped to a wrong position. For some cases, the regions near the border may lose the semantic pseudo labels as shown in Fig. 3. To alleviate these side-effects caused by the warping operation, we further rectify P'_{td} via a simple weighted fusion scheme using the nighttime prediction P_{tn}

$$F'_{td} = \eta P'_{td} + (1-\eta)P_{tn}, \tag{10}$$

where η is the confidence matrix which has the same shape as P'_{td} and the confidence value at each pixel i can be calculated by

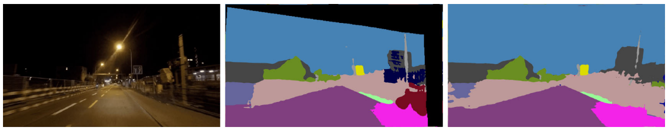


Fig. 4. The visualization of the pseudo label $\hat{F}_{td}^{S}$ without and with applying fusion scheme.

$$\eta(i) = \begin{cases} \eta_1 & \text{if } P'_{td}(i) > \theta, \\ \eta_2 & \text{otherwise.} \end{cases} \tag{11}$$

Here we set $\eta_1 = 0.8$, $\eta_2 = 0.6$, and $\theta = 0.4$, respectively. Our fusion operation is inspired by the confidence-adaptive prediction fusion proposed in [9], [19] which aims to refine the final dark image prediction based on its corresponding daytime prediction considering both of the dynamic and static contents of the two predictions. Specifically, the fusion in our static loss aims to refine the daytime prediction using its nighttime prediction to obtain a better pseudo label for training the network. Note that the fusion is a non-linear combination of P'_{td} and P_{tn} by applying Eq. (11). As shown in Fig. 4, the proposed fusion can rectify many wrongly warped labels, e.g., those on the buildings in this example.

We then apply Eq. (2) to calculate the re-weighted prediction $\hat{F}'_{td}$, and considering the static categories, the likelihood map $\hat{F}_{td}^{S} \in \mathbb{R}^{H \times W \times C^{S}}$ will be used to provide supervision.

We employ the focal loss [73] to remedy the imbalance among different categories of training samples. Finally, the static loss L_{static} is defined by

$$L_{static} = -\frac{1}{N} \| (1 - P_{tn}^{S})^{\gamma} log(p) \|_1, \tag{12}$$

where N is the total number of valid pixels in the segmentation ground truth, γ is the focusing parameter (set to 1 in all experiments), and p is the likelihood map for the correct category. Different from the focal loss in [73], we compute p at each pixel i in a 3×3 local region for category c by

$$p(c, i) = \max_j (o(c, j) \cdot P_{tn}^{S}(c, i)), \tag{13}$$

where o is the one-hot encoding of the semantic pseudo ground-truth which is derived from $\hat{F}_{td}^{S}$ by setting the maximum element to 1 and other elements to 0 for each pixel, and j represents each position of the 3×3 region centered at i. The pesudo label is still not perfectly aligned with the prediction after the warping and fusing operations. The usage of 3×3 region can help increase the fault tolerance rate and relieve the imperfection of the pseudo label.

3.5.4 Adversarial Loss

We employ two discriminators for adversarial learning, which are used to distinguish whether the output is from the source domain or one of the two target domains, i.e., S or T_d and S or T_n, respectively. We adopt the least-squares loss function [74] to make both predictions P_{td} and P_{tn} to be close to P_s. Specifically, we define the combination of these two adversarial losses (L_{adv}) as

$$L_{adv} = (D_d(P_{td}) - r)^2 + (D_n(P_{tn}) - r)^2, \tag{14}$$

where $P_{td} = G(I_{td})$, $P_{tn} = G(I_{tn})$, and r is the label for the source domain which has the same resolution as the output of discriminators [13]. Thus, the total loss L_{total} of the generator (G) is defined by combining L_{light}, L_{seg}, L_{static} and L_{adv}

$$\min_G L_{total} = \beta_1 L_{light} + \beta_2 L_{seg} + \beta_3 L_{static} + \beta_4 L_{adv}, \tag{15}$$

where $\beta_1, \beta_2, \beta_3$, and β_4 are set to 0.01, 1, 1 and 0.01 respectively in all experiments.

The generator and the corresponding discriminators are trained alternately and the objective functions of the discriminators D_s and D_n are defined respectively by

$$\min_{D_d} L_d = \frac{1}{2}(D_d(P_s) - r)^2 + \frac{1}{2}(D_d(P_{td}) - f)^2, \tag{16}$$

and

$$\min_{D_n} L_n = \frac{1}{2}(D_n(P_s) - r)^2 + \frac{1}{2}(D_n(P_{tn}) - f)^2, \tag{17}$$

where f is the label for the target domains with the same resolution as the output of discriminators.

4 Experiments

4.1 Datasets and Evaluation Metrics

For all experiments, we use the mean of category-wise intersection-over-union (mIoU) as the evaluation metric, and the higher the better. The following datasets are used for model training and performance evaluation:

Cityscapes [75]. The Cityscapes dataset contains 5,000 frames taken in street scenes with pixel-level annotations of a total of 19 categories, and both the original images and annotations have a resolution of $2,048 \times 1,024$ pixels. In total, there are 2,975 images for training, 500 images for validation and 1,525 images for testing. In this paper, we use the Cityscapes training set in the training stage of the proposed DANIA for adversarial learning.

Dark Zurich [9]. The Dark Zurich dataset consists of 2,416 nighttime images, 2,920 twilight images and 3,041 daytime images for training, which are all unlabeled with a resolution of $1,920 \times 1,080$. Images in these three domains can be coarsely aligned by using GPS-based nearest neighbor assignment to compensate the translation in each direction and the zoom in/out factors. In this paper, we only use 2,416 night-day image pairs in the training of the proposed DANIA (without using the twilight images). The Dark Zurich dataset also contains another 201 annotated nighttime images including 50 for validation (Dark Zurich-val) and 151 for testing (Dark Zurich-test), for quantitative evaluation. Note that Dark Zurich-test serves as an online benchmark whose ground truths are not publicly available. In our experiments, by submitting the segmentation results to the online evaluation website we get the performance of the proposed DANIA on Dark Zurich-test against the annotated ground-truths.

Nighttime Driving [18]. The Nighttime Driving test set contains 50 nighttime images of resolution $1,920 \times 1,080$ from diverse visual scenes. All these 50 images have been annotated at the pixel level using the same 19 Cityscapes category labels. In our experiments, we only use Nighttime Driving test set for performance evaluation without additional training.

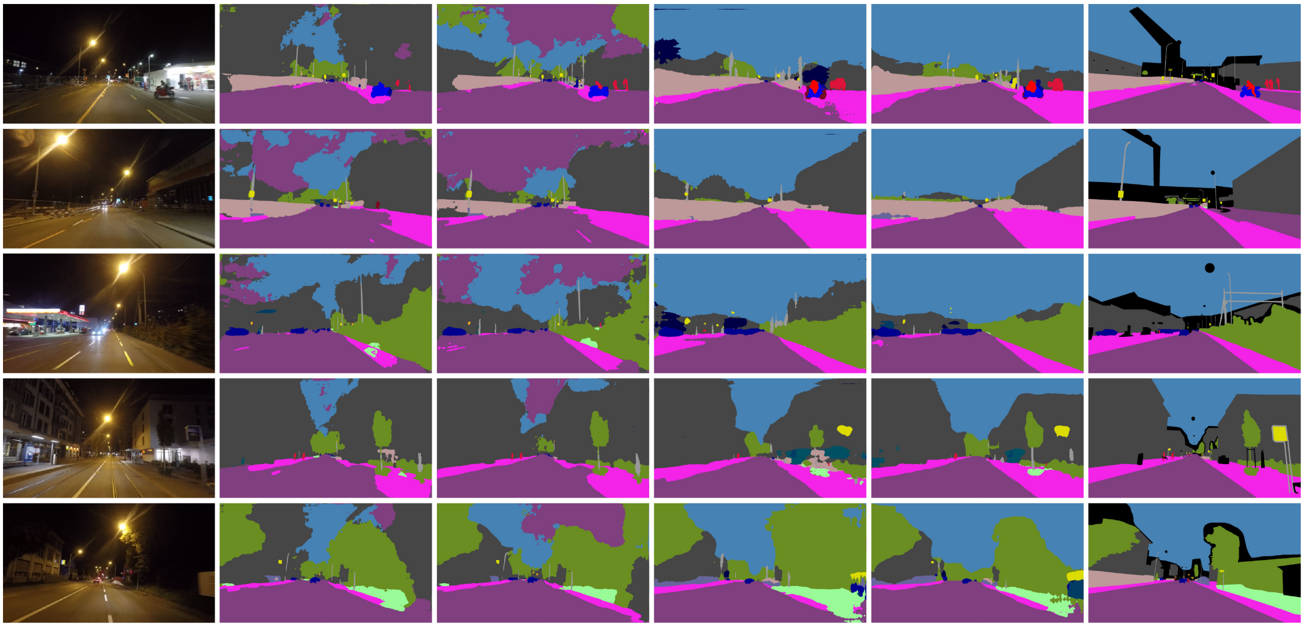


Fig. 5. Visualization comparison of our DANIA with some existing state-of-the-art methods on five samples from Dark Zurich-val.

ACDC [76]. ACDC is a newly released dataset which is introduced for semantic segmentation in adverse conditions. It contains 4,006 images in total including 1,000 foggy, 1,006 nighttime, 1,000 rainy and 1,000 snowy images. In particular, the images under each adverse condition is split into 400/100/500 for training/validation/test purpose respectively except for the nighttime domain, which has 106 validation images. Similar to Dark Zurich, each adverse condition image in ACDC comes with a corresponding image of the same scene that is taken in the normal conditions. The performance of our method on the ACDC-test is also obtained by submitting the segmentation results to an online server.[1]

4.2 Experimental Settings

We implement the proposed DANIA using PyTorch on a single NVIDIA Tesla V100 16GB GPU. Following [58], we train our network using the Stochastic Gradient Descent (SGD) optimizer with a momentum of 0.9 and a weight decay of 5×10^{-4}. The base learning rate is set to 2.5×10^{-4} and then we employ the poly learning rate policy to decrease it with a power of 0.9. The batch size is set to 2, and the total number of training iterations is set to 50,000. We use Adam optimizer [77] for training the discriminators with the coefficients being set to (0.9,0.99). The learning rate of the discriminators is set to 2.5×10^{-4} and follows the same decay strategy as for the generator. In addition, we apply random cropping with the crop size of 512 on the scale between 0.5 and 1.0 for Cityscapes dataset, with the crop size of 960 on the scale between 0.9 and 1.1 on Dark Zurich dataset, and random horizontal flipping in the training. To make the training easier to converge, we use the semantic segmentation models that are pre-trained on Cityscapes for 150,000 iterations and report the performance of different segmentation models on the validation set of Cityscapes and Dark Zurich in Table 1.

1. ACDC website: https://acdc.vision.ee.ethz.ch/

4.3 Comparison With Existing State-of-the-Art Methods

Comparison on Dark Zurich-Test. We first compare our DANIA with some existing state-of-the-art methods, including MGCDA [19], GCMA [9], DMAda [18] and several other domain adaptation approaches [13], [14], [38], [78], [79] on Dark Zurich-test, and the resulting mIoU performances are reported in Table 2. Among these methods, MGCDA, GCMA, and DMAda share the same baseline RefineNet while the rest are based on Deeplab-v2 and they use the common ResNet-101 backbone [61] and the nighttime images in Dark Zurich-test as inputs during testing. Note that FDA [79] is only trained based on a single spectral domain size of 0.09 without multiple rounds of the self-supervised training. To make a fair comparison, we show the mIoU performance of the proposed method with three segmentation baselines. For evaluation of each version of our DANIA on Dark Zurich-test, we directly employ the model weights that are saved in the last iteration (50,000) and find the optimal *std* that can achieve the best mIoU on the Dark Zurich-val, which is 0.14 for PSPNet, 0.10 for RefineNet and 0.12 for DeepLab-v2.

It can be observed that all of the three variants of DANIA outperform the others significantly. The improvements of DANIA over DANNet using the same baseline indicates the effectiveness of the proposed new features in this work, particularly, PSPNet (1.8%), RefineNet (2.9%) and DeepLab-v2 (2.3%). Different from DANNet, which achieves the best performance using the baseline model of PSPNet, the new DANIA achieves the best performance using the baseline model of RefineNet on the Dark Zurich-test. We also find that both DANNet and DANIA handle the large day-to-night domain gap very well and significantly outperform other methods on quite a few categories, such as road, sidewalk, and sky. Five sample visualization results on Dark Zurich-val are shown in Fig. 5, where we observe that both GCMA [9] and MGCDA [19] wrongly predict some sky and sidewalk regions as road. With the image alignment, DANIA predicts

TABLE 3
Comparison of Our DANIA With Some Existing State-of-the-Art Methods on Nighttime Driving-Test [18]

Method	mIoU(%)
RefineNet [59]-Cityscapes	32.75
DeepLab-v2 [58]-Cityscapes	25.44
PSPNet [60]-Cityscapes	27.65
AdaptSegNet-Cityscapes→DZ-night [13]	34.5
ADVENT-Cityscapes→DZ-night [78]	34.7
BDL-Cityscapes→DZ-night [14]	34.7
IntraDA-Cityscapes→DZ-night [79]	33.5
FDA-Cityscapes→DZ-night [38]	36.9
DMAda [18]	36.1
GCMA [9]	45.6
MGCDA [19]	**49.4**
DANNet (RefineNet) [24]	42.36
DANNet (DeepLab-v2) [24]	44.98
DANNet (PSPNet) [24]	47.70
DANIA (RefineNet)	45.65
DANIA (DeepLab-v2)	47.05
DANIA (PSPNet)	48.38

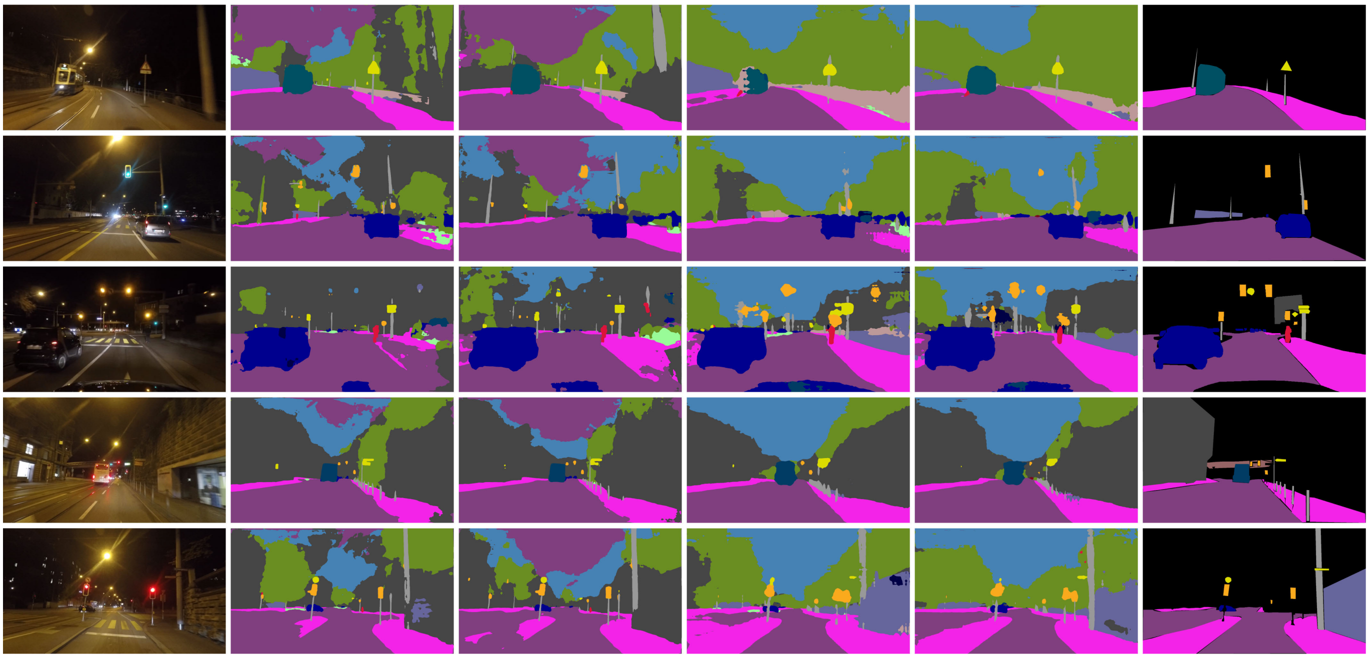
better along the boundaries of the objects such as the building (rows 1-4) and vegetation (rows 3-5).

Comparison on Night Driving. To show the generalization ability of the proposed DANIA, we also evaluate our models that are adapted to Dark Zurich on the test set of Night Driving. We report the performance of the proposed DANIA and the same set of comparison methods on Night Driving-test in Table 3, with some sample visualization results presented in Fig. 6. It is worth to mention that Night Driving dataset is not labeled as elaborately as Dark Zurich-test as shown in Fig. 6, where many categories that our DANIA predicts well (see Table 2), such as building and vegetation, are not annotated in this test set. We also notice that the category of sky is only labeled in 2 out of the 50 images in Night Driving test set. Even with these issues, our DANIA with PSPNet achieves the second best performance (MGCDA gets the best) on this dataset and surpasses the earlier DANNet using different baselines. These verify that our DANIA can generalize well to the Night Driving dataset. From Fig. 6, we observe that our methods give an appropriate prediction in the sky region while the others misclassify some sky pixels as road. Compared with DANNet, our DANIA did better on the train (row 1), traffic light (row 3) and sidewalk (row 5).

4.4 Ablation Study

Network Architecture. To demonstrate the effectiveness of each component of the proposed DANIA, we train several model variants for 50,000 iterations and use the model weights of the last iteration to test them on Dark Zurich-val. The performance results are reported in Table 4. Adaptation from Cityscapes to Dark Zurich-N using AdaptSegNet [13] serves as the baseline and DANIA is the full model.

Compared with the baseline and other two existing approaches [9], [19], our DANIA achieves impressive results on the Dark Zurich-val. We observe that coarsely aligned Dark Zurich-D is quite important although it is unlabeled, and the pseudo labels drawn from the predictions on Dark Zurich-D play a key role in our network, without which the mIoU decreases by 15.36%. Both the image relighting network and the corresponding loss L_{light} can enhance the performance and achieve 0.48% and 1.06% gains, although it only introduces 5.7% more parameters to the full model. To better illustrate the effects of relighting network, we visualize the light image ($I - R$) and the relighted image (R) in Fig. 7, where we observe that the illumination of the images from the three domains are adjusted. We also compute the intensity distribution of 500 images from three domains and their corresponding relighted images in Fig. 7, which indicates



Input image GCMA [9] MGCDA [19] DANNet (PSPNet) DANIA (PSPNet) Semantic GT

Fig. 6. Visualization comparison of our DANIA with some existing state-of-the-art methods on five samples from Night Driving-test.

TABLE 4
Ablation Study on Several Model Variants of Our DANIA (PSPNet) on Dark Zurich-val

Method	mIoU(%)
GCMA [9]	26.65
MGCDA [19]	26.10
AdaptSegNet-Cityscapes→DZ-night [13]	20.19
w/o Dark Zurich-D	22.78
w/o relighting network & L_{light}	37.66
w/o L_{light}	37.08
w/o pretrained segmentation model	19.33
DANIA	**38.14**

TABLE 5
Ablation Study for Static Loss L_{static} in Our DANIA (PSPNet) on Dark Zurich-val

Method	mIoU(%)
w/o L_{static}	22.50
w/ Cross Entropy Loss in L_{static}	34.46
w/ Weighted Cross Entropy Loss in L_{static}	34.61
w/ Focal Loss in L_{static}	33.92
w/o prediction fusion	33.97
w/o re-weighting on pseudo labels	37.05
w/o re-weighting on prediction	35.42
DANIA	**38.14**

that the relighted intensity distribution of the three domains gets much closer to each other.

Static Loss and Re-Weighting Strategy. We also conduct ablation studies to verify the effectiveness of the static loss L_{static} and the re-weighting strategy. The quantitative results are presented in Table 5. We can see that the specially designed loss L_{static} with the 3×3 local-region design is better than directly applying the cross entropy, weighted cross entropy or focal loss to calculate the static loss. In addition, the fusion scheme is verified to be useful and can further boost the performance.

The re-weighting strategy can also enhance the semantic segmentation performance and achieve 1.09% gains when applying to the pseudo labels, and 1.06% gains when applying to the final predictions. As shown in Fig. 8, this strategy helps segment the small objects. We find that the selection of the value std is also important in applying the re-weighting strategy. We test different std values and the performance curve of the proposed DANIA on Dark Zurich-val is shown in Fig. 9. The optimal performance for DANIA (PSPNet) is achieved with $std = 0.14$ during evaluation.

Hyperparameter. Our method contains several hyperparameters especially in the loss functions and the fusing operation. To learn the sensitivity of our method to the choice of hyperparameters, we train our model with multiple group of values. We first study the different choices of weights in Eqs. (7) and (15) as shown in Table 6. We observe that the model is not sensitive to the weights in Eq. (7) but it is quite sensitive to weights β_1 and β_4 in Eq. (15). In Table 7, we also study the parameter values in the fusing operation. We find that the performance is relatively stable when changing the parameters in Eq. (11).

Image Alignment. We also try to use the image alignment method in different ways to establish the correspondence between the day-night pairs. One is achieved by warping the daytime prediction using the estimated coarse flow

$$P'_{td} = \mathcal{F}_{coarse}(P_{td}). \tag{18}$$

The second is achieved by warping the daytime images using the estimated fine flow instead of warping the daytime prediction

$$I'_{td} = \mathcal{F}_{fine}(I_{td}). \tag{19}$$

The last one is to warp the nighttime predictions P_{tn} using the fine flow

$$P'_{tn} = \mathcal{F}^{-1}_{fine}(P_{tn}), \tag{20}$$

where $\mathcal{F}^{-1}_{fine}$ represent the estimated fine flow from I_{tn} to I_{td}.

The results of all the alternatives of image alignment method on Dark Zurich-val are reported in Table 8, where

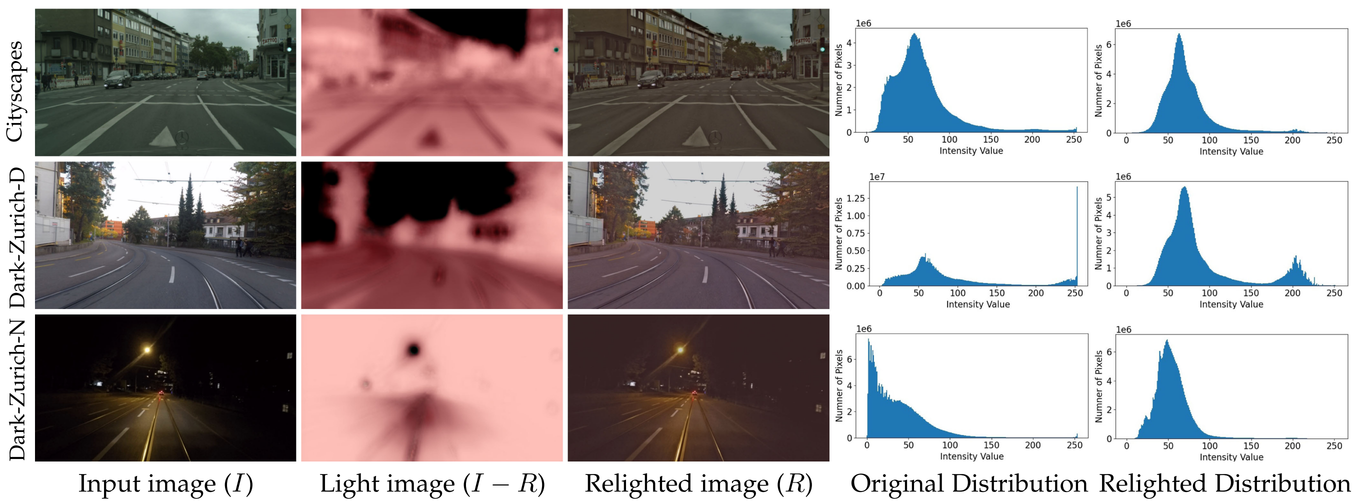


Fig. 7. Visualization of relighted images from the three domains. From top to bottom, we show the three samples from the Cityscapes, Dark Zurich-D and Dark Zurich-N, respectively. The original intensity distribution is calculated using 500 input images from each domain (the fourth column) and the relighted intensity distribution is calculated from 500 corresponding relighted images (the last column).

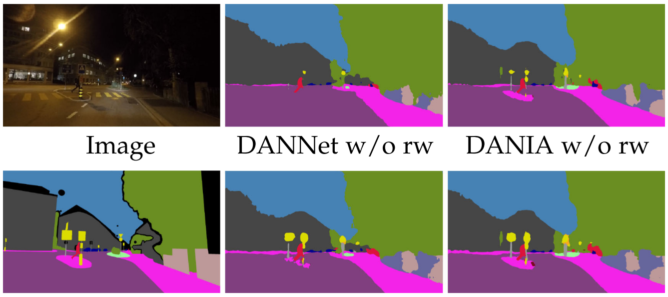


Fig. 8. Visualization results of w/ and w/o the re-weighting (rw) strategy on a sample from Dark Zurich-val by the DANNet (PSPNet) and DANIA (PSPNet).

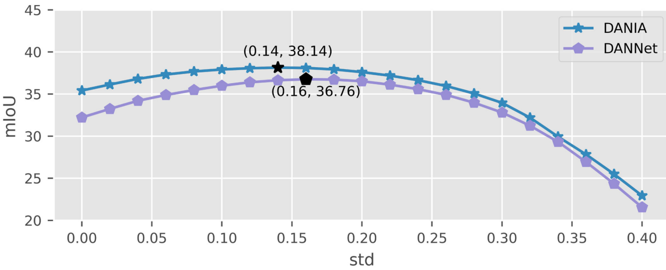


Fig. 9. Ablation study on the value of std in the re-weighting strategy on Dark Zurich-val by our DANIA (PSPNet) and DANNet (PSPNet).

TABLE 6
Ablation Study for Loss Weights in Eqs. (7) and (15) Based on DANIA (PSPNet) on Dark Zurich-val

weights	mIoU(%)
$\alpha_{tv}, \alpha_{exp}, \alpha_{style}, \alpha_{ssim} = 1, 1, 1, 1$	36.64
$\alpha_{tv}, \alpha_{exp}, \alpha_{style}, \alpha_{ssim} = 10, 1, 1, 1$	36.47
$\alpha_{tv}, \alpha_{exp}, \alpha_{style}, \alpha_{ssim} = 1, 1, 1, 10$	37.88
$\alpha_{tv}, \alpha_{exp}, \alpha_{style}, \alpha_{ssim} = 10, 1, 10, 1$	**38.14**
$\beta_1, \beta_2, \beta_3, \beta_4 = 1, 1, 1, 1$	15.45
$\beta_1, \beta_2, \beta_3, \beta_4 = 0.1, 1, 1, 1$	9.53
$\beta_1, \beta_2, \beta_3, \beta_4 = 1, 1, 1, 0.1$	37.17
$\beta_1, \beta_2, \beta_3, \beta_4 = 0.1, 1, 1, 0.1$	37.70
$\beta_1, \beta_2, \beta_3, \beta_4 = 0.01, 1, 1, 1$	10.61
$\beta_1, \beta_2, \beta_3, \beta_4 = 1, 1, 1, 0.01$	37.49
$\beta_1, \beta_2, \beta_3, \beta_4 = 1, 1, 1, 0.001$	36.59
$\beta_1, \beta_2, \beta_3, \beta_4 = 0.01, 1, 1, 0.01$	**38.14**

we find that directly applying the coarse flow (homography transform) can bring in 2.93% mIoU gains. By using the fine flow, we further achieve additional 0.18% on mIoU, which is not a big improvement. We think the reason is that fine flow model suffers from a domain gap in our framework since it was not trained on our used dataset. We also notice that both Eqs. (19) and (20) only improve about 0.3% on mIoU. While warping either the daytime image or the prediction may introduce holes in the results, the holes in the input image produce wrong daytime prediction and affect the calculation in the following steps. On the contrary, the holes in the warped prediction can be simply treated as an ignored class.

4.5 Day-Night Pairs at Test Time

For all of the above experiments that are related to ours, the day-night pairs are used for network training. During testing, only the nighttime image is processed by the network to make the prediction. This is totally reasonable since the day-night pairs may not be available for testing in many real-time application. However, in general cases, the daytime counterpart is achievable with additional effort. Therefore, we also investigate the effectiveness of using paired data during testing and the results are reported in Table 9.

TABLE 7
Ablation Study for Parameters in Eq. (11) Based on DANIA (PSPNet) on Dark Zurich-val

parameters	mIoU(%)
$\eta_1, \eta_2, \theta = 1.0, 0.8, 0.2$	37.96
$\eta_1, \eta_2, \theta = 0.6, 0.4, 0.2$	37.40
$\eta_1, \eta_2, \theta = 0.8, 0.2, 0.4$	37.85
$\eta_1, \eta_2, \theta = 0.8, 0.4, 0.4$	36.85
$\eta_1, \eta_2, \theta = 0.8, 0.6, 0.2$	36.65
$\eta_1, \eta_2, \theta = 0.8, 0.6, 0.6$	36.84
$\eta_1, \eta_2, \theta = 0.8, 0.6, 0.4$	**38.14**

TABLE 8
Ablation Study for Image Alignment Our DANIA (PSPNet) on Dark Zurich-val

Method	mIoU(%)
without using image alignment	35.03
Using Eq. (18)	37.96
Using Eq. (19)	35.33
Using Eq. (20)	35.34
DANIA	**38.14**

TABLE 9
Comparison With the Usage of Day-Night Pairs on Dark Zurich-Test According to mIoU(%)

Method	night	day-night pair
DMAda [18]	32.1	34.8
MGCDA [19]	42.5	44.1
DANNet (PSPNet) [24]	45.2	45.5
DANIA (DeepLab-v2)	44.8	46.1
DANIA (PSPNet)	47.0	47.7
DANIA (RefineNet)	47.2	**47.8**

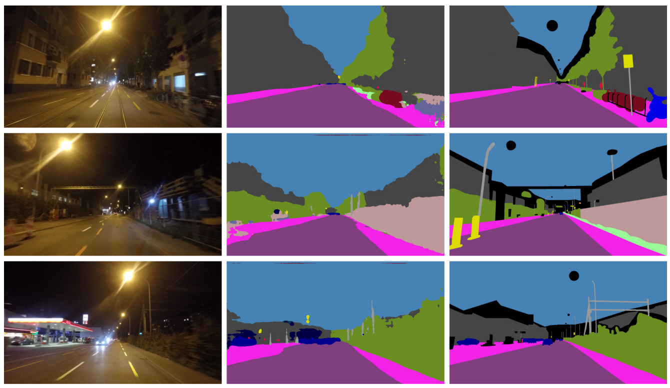


Fig. 10. The visualization results of three failure cases from Dark-Zurich Val by DANIA (PSPNet).

TABLE 10
Comparison Results of the Usage of Deblur and Image Enhancement Approaches on Dark Zurich-val

Method	mIoU(%)
ours	38.14
w/ deblur [80]	36.82
w/ image enhancement [67]	30.56

In order to make use of the daytime images, we follow our network training step by first estimating the fine flow and use it to perform a daytime prediction warping. Then an adjusted nighttime prediction can be achieved with the guidance of the warped daytime prediction via the fusion scheme. From the results, we can see that the daytime images can help segment their nighttime counterpart for better segmentation accuracy in the testing.

4.6 Failure Cases

In Fig. 10, we show three failure cases produced by our DANIA (PSPNet). The wrong prediction appears in the regions that contain large magnitude of motion blur (e.g., the bicycles and motorcycle in the first row and the traffic signs in the second row) or artificial lights (e.g., the lights from gas station in the third row). Therefore, we further consider to

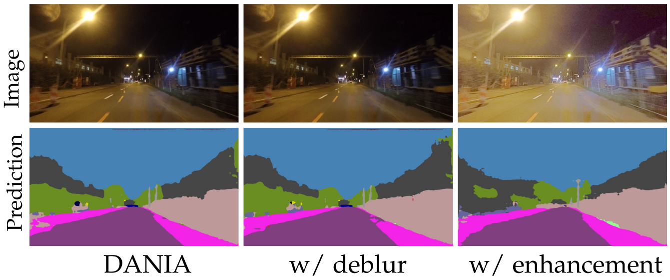


Fig. 11. Visualization results of using image pre-processing approaches on Dark Zurich-val. The original image and its ground-truth segmentation are shown in the second row of Fig. 10.

apply a deblur approach [80] and an image enhancement approach [67] on the Dark Zurich-test to adjust the blurring effects and the illumination of the images as a pre-processing stage.

The quantitative results of with and without these two image pre-processing approaches are reported in Table 10, where we observe that the use of these two pre-processing approaches suffer from the performance drop. A visual comparison is also provided in Fig. 11. The image pre-processing approaches did not make any improvements for the wrong predictions. We find that the deblurring effect is not very obvious on the nighttime images which indicates that

TABLE 11
The Per-Category mIoU (%) on ACDC Dataset by GCMA [9], MGCDA [19] and Our Proposed DANIA

Subset	Method	road	sidewalk	building	wall	fence	pole	traffic light	traffic sign	vegetation	terrain	sky	person	rider	car	truck	bus	train	motorcycle	bicycle	**mIoU**
All	GCMA [9]	79.7	48.7	71.5	21.6	29.9	**42.5**	**56.7**	**57.7**	75.8	39.5	87.2	**57.4**	29.7	**80.6**	44.9	**46.2**	62.0	**37.2**	**46.5**	53.4
	MGCDA [19]	76.0	49.4	72.0	11.3	21.7	39.5	52.1	54.9	73.7	24.7	88.6	54.2	27.2	78.2	30.9	41.9	58.2	31.1	44.4	48.9
	DANIA (DeepLab-v2)	87.8	57.1	80.3	36.2	31.4	28.6	49.5	45.8	76.2	48.8	90.2	47.9	**31.1**	75.5	36.5	36.5	47.8	32.5	44.1	51.8
	DANIA (PSPNet)	88.4	**60.6**	81.1	37.1	32.8	28.4	43.2	42.6	77.7	50.5	90.5	51.5	**31.1**	76.0	37.4	44.9	64.0	31.8	46.3	53.5
	DANIA (RefineNet)	**89.4**	59.8	**82.1**	**39.5**	**36.0**	28.9	45.1	44.9	**80.3**	**53.6**	**92.1**	49.9	28.2	79.5	**49.4**	41.4	**64.6**	34.3	44.7	**54.9**
Night	GCMA [9]	78.6	45.9	58.5	17.7	18.6	37.5	**43.6**	**43.5**	58.7	**39.2**	22.5	**57.9**	29.9	**72.1**	21.5	**56.3**	41.8	35.7	35.4	42.9
	MGCDA [19]	74.5	52.5	69.4	7.7	10.8	**38.4**	40.2	43.3	61.5	36.3	37.6	55.3	25.6	71.2	10.9	46.4	32.6	27.3	33.8	40.8
	DANIA (DeepLab-v2)	89.4	55.1	75.2	37.5	27.1	33.1	37.0	37.2	70.5	34.2	78.9	47.8	32.4	64.9	6.1	34.8	41.7	27.3	36.2	45.6
	DANIA (PSPNet)	91.0	60.9	77.7	**40.3**	**30.7**	34.3	37.9	34.5	70.0	37.2	79.6	45.7	**32.6**	66.4	11.1	37.0	**60.7**	32.6	**37.9**	48.3
	DANIA (RefineNet)	**92.1**	**63.0**	**79.0**	39.3	**30.7**	33.1	39.3	40.2	**70.7**	37.5	**80.6**	49.1	28.9	71.5	**48.3**	56.0	59.0	**43.3**	36.9	**52.6**
Fog	GCMA [9]	80.8	53.6	70.1	29.2	20.7	**38.4**	**53.0**	**60.9**	70.2	46.5	95.4	44.2	38.0	**76.6**	52.4	49.7	56.8	**41.0**	17.6	**52.4**
	MGCDA [19]	71.7	47.3	65.7	18.2	15.3	34.4	48.6	59.9	64.9	24.7	95.4	**44.8**	23.8	73.3	36.1	45.4	**63.9**	24.0	15.4	45.9
	DANIA (DeepLab-v2)	**85.8**	**55.9**	77.4	26.1	24.6	25.3	41.6	44.5	79.1	56.1	**95.5**	28.1	41.1	67.7	36.8	37.9	57.0	23.8	19.0	48.6
	DANIA (PSPNet)	85.4	50.3	**80.8**	**36.1**	25.5	26.9	39.8	45.4	**79.7**	**60.2**	95.3	34.6	**46.2**	69.9	48.2	55.3	55.9	4.8	**29.0**	51.0
	DANIA (RefineNet)	84.4	50.1	77.7	35.5	**28.5**	26.8	33.7	2.3	78.5	59.7	88.5	33.2	22.7	75.5	**53.8**	**73.5**	56.3	28.5	26.9	49.3
Rain	GCMA [9]	81.1	48.0	84.8	25.0	37.3	**49.8**	**66.5**	**66.2**	**92.1**	43.5	**97.6**	**54.5**	20.4	**85.5**	47.3	34.6	**71.3**	**40.3**	**56.7**	**58.0**
	MGCDA [19]	80.5	46.6	79.9	16.0	28.8	44.9	60.1	61.5	90.3	**44.8**	97.1	51.1	**23.1**	82.3	33.4	30.2	69.1	36.5	53.8	54.2
	DANIA (DeepLab-v2)	84.9	56.3	83.0	39.0	37.0	25.4	25.9	44.5	86.2	36.7	93.1	42.3	12.3	79.6	36.8	13.9	56.1	29.7	49.1	49.0
	DANIA (PSPNet)	85.8	60.6	84.2	47.9	40.1	29.4	43.8	44.1	88.2	39.4	91.1	44.7	16.5	82.8	**61.4**	55.0	62.6	37.1	53.4	56.2
	DANIA (RefineNet)	**87.3**	**61.4**	**86.5**	**50.7**	**40.4**	34.2	42.1	45.2	88.5	37.0	95.5	44.0	14.8	84.8	59.1	**55.8**	59.9	32.1	48.5	56.2
Snow	GCMA [9]	79.7	49.6	75.3	17.5	**37.9**	**43.2**	**59.0**	**61.9**	78.8	2.2	**95.5**	**62.5**	33.6	**83.2**	42.5	43.4	72.1	**32.2**	51.1	53.8
	MGCDA [19]	80.1	49.5	70.2	6.1	27.8	39.6	55.4	58.0	76.0	0.3	**95.5**	57.5	35.7	81.0	28.6	**48.9**	70.3	27.8	50.5	50.5
	DANIA (DeepLab-v2)	83.9	52.3	77.4	23.0	30.7	31.5	51.7	49.1	77.9	22.4	90.8	42.8	25.6	78.6	30.8	33.5	64.0	23.0	41.8	49.0
	DANIA (PSPNet)	85.3	**56.7**	79.2	30.0	34.9	31.8	53.8	44.1	78.8	25.1	90.5	57.1	34.3	80.1	40.7	45.9	63.4	26.3	51.4	53.1
	DANIA (RefineNet)	**85.6**	54.2	**80.8**	**32.3**	32.6	32.8	45.7	47.1	**81.7**	**30.7**	92.0	57.3	**46.3**	81.4	**43.0**	46.9	**70.8**	27.4	**52.0**	**57.8**

The best results are presented in bold. *Note that the performance on all conditions is not an average of the performances of the four conditions.*

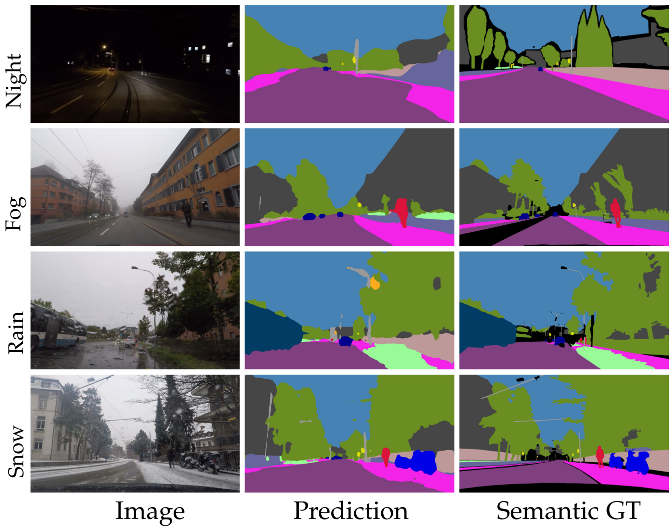


Fig. 12. Visualization results of samples in ACDC dataset by DANIA (RefineNet).

the deblur algorithm [80] may not generalize well to the Dark Zurich images. Although the image enhancement preprocessing [67] makes the image brighter, it may lead to more domain shift between the training and test images and hurt the segmentation accuracy.

4.7 Experiments on the ACDC Dataset

We further explore our method on the newly released ACDC dataset which contains not only a larger scale of nighttime dataset than the Dark Zurich but also image pairs in other adverse conditions including fog, rain and snow. We train our model separately for each adverse condition. The quantitative results are shown in Table 11. We observe that our method with RefineNet as the backbone achieves the best performance on the night, snow and "all" conditions and the second best on the fog and rain conditions. Note that the performance on all conditions is not an average of the performances of the four conditions. To achieve the performance for "all" condition, we train our model with 2,000 image pairs from all of the four conditions. Sample qualitative results are shown in Fig. 12.

5 Conclusion

In this paper, we have proposed a novel end-to-end neural network DANIA for unsupervised nighttime semantic segmentation, which performs a one-stage adaptation from a labeled daytime dataset to unlabeled day-night image pairs. In the proposed DANIA, an image relighting network with a special light loss function is first used to make the intensity distributions of the images from different domains to be close to each other. The unlabeled Dark Zurich-D data is used to bridge the domain gap between the labeled daytime images (Cityscapes) and the unlabeled nighttime images (Dark Zurich-N). In addition, an image alignment step is incorporated to address the misalignment between the day-night image pairs. By leveraging the similar illumination patterns between Dark Zurich-D and Cityscapes and the adjusted alignment of static object categories between Dark Zurich-D and Dark Zurich-N, our DANIA performs multi-target domain adaptation and a re-weighting strategy to boost the performance for small objects. Experimental results demonstrated the effectiveness of each of the designed components and showed that our DANIA achieves the state-of-the-art performance on Dark-Zurich and Night Driving test datasets.

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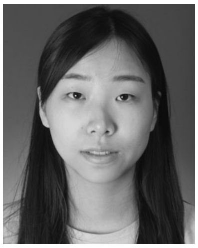

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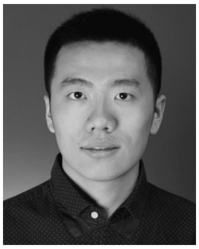

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